UTK EcoCAR Year-1 Projects

Relevance Assessment for Electrical Engineering,
Computer Engineering, and Computer Science Undergraduates

Scoring Scale

Score	Meaning
5	Core to the major — draws directly on required coursework
4	Strong fit — substantial overlap with major coursework
3	Moderate fit — meaningful contribution possible, some ramp-up required
2	Limited fit — peripheral technical overlap
1	Little to no technical overlap with the major

Grading Summary

#	Project	EE	CpE	CS	Overall	Rationale
1	CAD Design & Data Management	1	2	3	Low	Mostly mechanical CAD, but the PLM/Teamcenter work is really configuration control, metadata, and release workflows — a CS student comfortable with version control and data schemas fits better here than an EE.
2	Off-Road Performance Target Development	1	1	1	Very Low	Ground clearance, suspension travel, approach/breakover/departure angles — almost purely mechanical and vehicle dynamics.
3	PCM Tool Check & Controls Development Readiness	5	5	4	**Very High**	Simulink toolchain, Git-based version control, MIL infrastructure, and software development workflows. This is the software-engineering backbone of the propulsion controls swimlane.
4	Battery Performance Target Development	5	3	2	High (EE-anchored)	Power, current, voltage, and energy sizing plus cell/module configuration is squarely EE. CpE contributes on the BMS and data side; limited CS pull.
5	Virtual Vehicle Simulation	4	4	4	High	MATLAB/Simulink modeling, EPA drive-cycle analysis, torque split, and propulsion-control behavior. Modeling and data analysis are accessible to all three majors.

#	Project	EE	CpE	CS	Overall	Rationale
6	Final Technical Report / IEEE ITEC Paper	5	4	3	High	ITEC is a transportation *electrification* conference — the natural topics (controls, power electronics, energy management, embedded software) are ECE territory. CS fit depends entirely on topic selection.
7	Integrated Vehicle Architecture Selection	4	3	3	Moderate–High	Systems engineering, trade studies, requirements traceability, and preliminary hazard analysis. Strong for a systems-minded student; less hands-on circuit or code work.
8	Baseline Vehicle Testing & Characterization	5	5	3	**Very High**	Instrumentation, data acquisition, CAN bus communications, sensor signal chains, and battery behavior. Likely the best hands-on entry point for ECE underclassmen.
9	SDV Ecosystem, Market Analysis & Wedge Discovery	2	4	4	Moderate	Software-defined vehicle domain literacy is a CS/CpE strength — identifying technical bottlenecks requires understanding automotive software stacks. The deliverable, however, is market analysis rather than code.
10	Customer Discovery & Product Innovation Validation	1	1	2	Low	Interviews, business model updates, and Desirability–Feasibility–Viability analysis. Technical background helps only in interpreting responses.
11	Mobility Problem-Space Exploration	1	1	2	Low	Observation and research with no engineering deliverable.
12	Integrated Project Management & Digital Project Controls	1	2	3	Low–Moderate	WBS and scheduling are discipline-agnostic, but "digital project controls" and status dashboards involve real tooling a CS student could build.
13	University Stakeholder Engagement	1	1	1	Very Low	Executive outreach, briefings, and communications.

Recruiting Tiers

Tier 1 — Lead with these

Projects **3, 8, 5, 4, and 6**. These map directly to the departmental skill inventory: MATLAB/Simulink, sensor instrumentation and data acquisition, battery/electrochemical systems, electric and hybrid vehicles, and computer programming/scripting.

Tier 2 — Good fit for specific student profiles

- **Project 7** — students with an interest in systems engineering and requirements work
- **Project 9** — CS students seeking industry exposure without a mechanical prerequisite

Tier 3 — Fill primarily with mechanical engineering or business majors

Projects **2, 10, 11, and 13**.

Notes and Recommendations

Project 9 is the most likely to be undersold. The title reads like a business project, but "technical bottlenecks in the software-defined vehicle ecosystem" is a topic where a CS student holds an advantage over everyone else on the team. Position it accordingly in recruiting materials.

Project 1 should be reframed if CS interest is desired. Describe it as data management and change control rather than CAD — the CAD-authoring skill barrier is what will deter CS students from a project where their actual contribution would be workflow and configuration management.

Project 6 requires early topic scoping. Its EE relevance is high by default, but a CS student's fit swings between a 2 and a 5 depending on whether the chosen research topic leans toward power electronics or toward controls software and modeling.

Skill Mapping Reference

Departmental skill areas with the strongest project coverage:

Skill Area	**Best-Matched Projects**
MATLAB/Simulink	3, 5, 7
Numerical Modeling / Simulation	5, 7, 4
Battery / Electrochemical Systems	4, 8
Electric / Hybrid Vehicles	4, 5, 6, 7
Vehicle Dynamics and Control Systems	3, 5, 7
Sensor Instrumentation + Data Acquisition	8
Experimental Testing	8, 4
Computer Programming / Scripting	3, 5, 9, 12
System Engineering	7, 3, 12
Project Management	12, 1